Lesson notes · for you

How machines learn

AI · Level 1 · Lesson 2

Time
25 min

Materials
Paper and a pen; A friend or family member to play with; Squared paper, if you have some

Learning goals

- Put the five steps of machine learning in order.
- Explain why a model is tested on examples it has never seen.
- Explain why more, and more varied, examples help.

Activity: Be the machine

1. Choose (3 min): Pick a secret pattern that is easy to draw, such as “has a dot inside”, “has exactly three corners” or “is shaded”. Write it down and keep it hidden.
2. Train (6 min): Draw 8 shapes that follow the pattern under the label “yes” and 8 that don’t under “no”. Vary size, direction and shading in both groups, so only the pattern tells them apart. Your partner studies them for 2 minutes: this is the training data.
3. Test (6 min): Hide the training drawings. Draw 6 new shapes one by one, including one or two tricky ones. Your partner says “yes” or “no” for each: this is the test on examples they have never seen.
4. Score (4 min): Count the right answers out of 6, then reveal the pattern. Which shape fooled them, and which extra training example would have helped?
5. Swap and talk (6 min): Swap roles and play again. Then name the five steps you used: collect, label, train, test, use. Say why testing with the same 16 drawings would have been too easy.

You need paper and a pen for each pair. Plan the secret pattern and the 6 test shapes before you start, and make sure at least one test shape looks different from all the training shapes.

Discussion

- What would you teach a machine to sort, and what labels would you use?
- Why is memorising last year’s exam paper a risky way to revise?
- Whose messages, voices or photos might be missing from examples collected only at your school?

Check yourself

- Write the five steps in order, and one reason to test on new examples.

Try it this week
Play “Be the machine” with a friend or someone at home: draw shapes that follow a secret pattern and shapes that don’t, let them find the pattern, then test them with new shapes.

Worksheet

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Name _____ Class _____ Date _____

1 **A scam filter learned only from messages promising free data. A fake scholarship offer arrives. What's most likely?**

◯ It catches it easily ◯ It learns the new trick by itself straight away ◯ It might miss it

2 **Put the five steps in order.**

☐ Use it on new messages

☐ Test on examples it never saw

☐ Train: find the patterns

☐ Label each one

☐ Collect examples

3 **Your filter scores 18 out of 20 on the messages you kept aside. That tells you more than 20 out of 20 on its training messages would.**

◯ True ◯ False

4 **Complete the sentence.**

We _____ each example with the right answer, _____ the model to find patterns, then _____ it on examples it has never seen.

Word bank: label · train · test · memorise · delete

✎ **What would you like to teach a machine to sort, at school or at home? Which examples and labels would it need?**